

JOR: A Domain-Agnostic Framework for Structured Uncertainty

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Purpose

Provide a transparent, conservative, reproducible method for evaluating ambiguous, incomplete, or conflicting observations across any industry.

Core Problems JOR Solves

- **Ambiguous data** — inconsistent, partial, or low-quality inputs
- **Uncertainty quantification** — determining confidence levels
- **Evidence fusion** — combining multiple sources without runaway certainty
- **Reproducible scoring** — identical inputs produce identical outputs
- **Bounded reasoning** — explicit caps prevent false confidence
- **High-volume triage** — ranking millions of observations quickly (Note: Refers to JOR’s ability to process millions of structured observations; not connected to live sensors.)

JOR achieves this using: Conservative Bayesian math • Explicit uncertainty caps • Vectorized ingestion • Transparent logic • Domain-agnostic architecture

Cross-Industry Applications

Aviation (Analytic Use Only)

- Pre-flight anomaly triage
- Post-incident review
- Human-report uncertainty scoring
- Conflicting-sensor interpretation
- Data-quality assessment

Example: A safety analyst structures a pilot report and uses JOR to generate a reproducible SOP posture for pre-flight awareness.

Defense & National Security

- Ambiguous-event triage
- Multi-source uncertainty fusion
- Structured analytic techniques
- Wargaming uncertainty modeling

Example: Scenario designers score ambiguous intel packets with JOR to inject realistic uncertainty into tabletop decision-making.

Research & Academia

- Evidence weighting
- Cross-study comparison
- Methodological transparency

Example: Researchers score evidence quality across multiple studies to create a transparent, reproducible weighting scheme.

Commercial Analytics

- Customer-report triage
- Fraud-signal scoring
- Quality-control anomaly detection

Example: A product team uses JOR to score thousands of customer-reported anomalies and surface the most credible issues for engineering review.

Engineering & Safety-Critical Fields

- Failure-mode triage
- Structured incident review
- Safety case documentation

Example: Engineers score ambiguous equipment failures with JOR to identify which incidents warrant deeper investigation.

Sensor Integration & Design Intent

Current State: JOR is a standalone analytic framework. It does not ingest live sensor data or connect to operational systems.

Design Intent: Preserves transparency, reproducibility, and domain-agnostic flexibility. JOR logic is open-source and fully auditable.

Future Compatibility

JOR is architecturally compatible with sensor pipelines, and could be connected to live sensors in future implementations, but the current framework relies entirely on structured/manual case scoring. JOR can integrate with sensor-driven pipelines because it operates on structured observations, not hardware. It can accept: Radar-derived summaries, ADS-B / FOQA event flags, telemetry-based observations, human-reported observations, and fused or pre-processed sensor outputs.

JOR does not require sensors — but it is fully compatible with them.

Why JOR Works Everywhere

Methodology: <https://doi.org/10.5281/zenodo.18088931>

Every industry faces ambiguous data, conflicting reports, incomplete information, human judgment, and uncertainty that must be bounded.

JOR provides: Transparency • Reproducibility • Conservative math • Scalable performance • Domain-agnostic logic.